Complete Java Classes

by Kunal Sir

Has-A Relationship in Java:

Has-A relationship in java is known to be as composition.

It is used for code reusability.

Basically, it means that an instance of the one class has a reference to the instance of another class or the other instance of the same class.

This relationship helps to minimize the duplication of code.

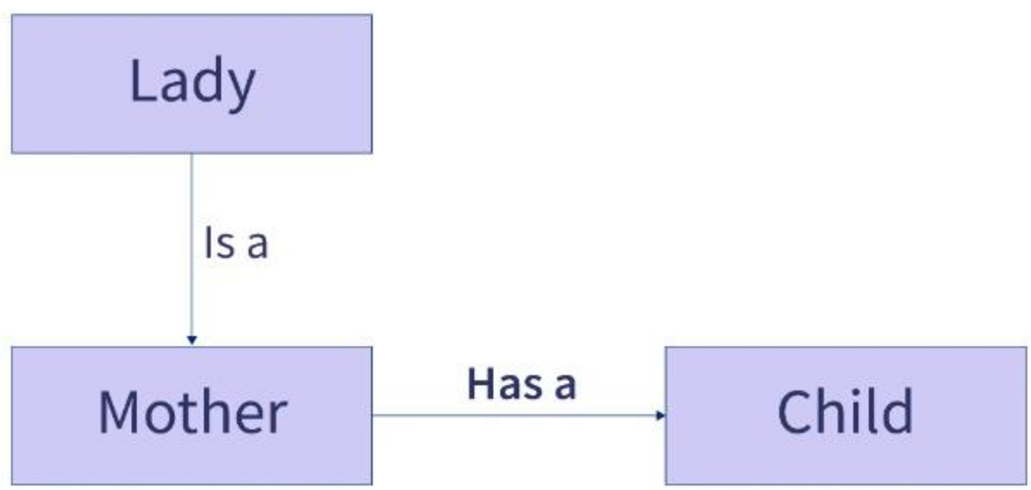
Composition and Aggregation are the two forms of association.

Association: - Association is the relation between two separate classes which establishes through their Objects.

Example of Has-A Relationship:

Consider a lady, she can be called a mother only when she has a child. And a child cannot exist without their mother.

We can understand the below example by the following diagram-



In the above diagram, we can see that the mother class is derived from the lady class. So, this type of relationship is called an Is-A relationship. But our main focus is on the **Has-A Relationship** that exists between mother and child. These two classes are tightly coupled, which means that the existence of one class depends on another class. If the child does not exist, then a lady cannot be called a mother and vice versa.

Understanding Program:

```
public class Student
{
  int sid;
  String sname;

   public Student(int sid, String sname)
   {
        this.sid = sid;
        this.sname = sname
   }
}


public class School
{
  int schid;
  String schname;
  Student stu;                   // Has-A Relationship

   public School(int schid, String schname, Student stu)
   {
        this.schid = schid;
        this.schname = schname;
        this.stu = stu;
   }
}


public class College
{
  int cid;
```

```
  String cname;
  School sch;                    // Has-A Relationship

  public College(int cid, String cname, School sch)
  {
      this.cid = cid;
      this.cname = cname;
      this.sch = sch;
  }
}


public class Test
{
  public static void main(String args[ ])
  {
      Student s = new Student(1, "Ram");

      School sc = new School(15, "DPS", s);

      College c = new College(25, "FC", sc);

      System.out.println("College Id = "+c.id);
      System.out.println("College Name = "+c.cname);
      System.out.println("School Id = "+c.sch.schid);
      System.out.println("School Name = "+c.sch.schname);
      System.out.println("Student Id = "+c.sch.stu.sid);
      System.out.println("Student Name = "+c.sch.stu.sname);
   }

}
```